8.2 Binomial Nomenclature

Swedish botanist Carolus Linnaeus in 1735 introduced a formal system of classification which divided all living organism into two kingdoms-Animalia and Plantae. He introduced "two name" system, the first name, genus and second name species. The name often gives information on something special about it. Taxa (the basic taxonomic group) are constructed from strains which are successions of cultures derived from an initial colony. The basic taxonomic group is called the species (a collection of strains having similar characteristics). The special bacterial strain which is the permanent reference specimen for the species is called the "**type strain**" (Figure 8.2).

HOTS

Why is type strain refered as the most important strain in a bacterial species?

A variant strain that differ physiologically and biologically from other strains in a particular species is called as "Biovar". Variations in a species is biological in nature. One biovar in a species may grow on sucrose, while another cannot. If the biovars are very similar except for one property, they belong to the same genus and species, though vary in biological growth properties.

The strain that differ morphologically are called as Morphovar or Morphotypes. Serovars or Serotypes are those strains that differ in their antigenic properties. It refers to immunological variations in a species. An example of differing serovars is *Salmonella*. Cell surface of *Salmonella* varies slightly from one serovar to another. Because of this cell surface change, a person who has been infected by or become resistant or immune to one serovar will not be immune to a second type, because the immune system cannot recognize a similar bacterium with a new surface cover.

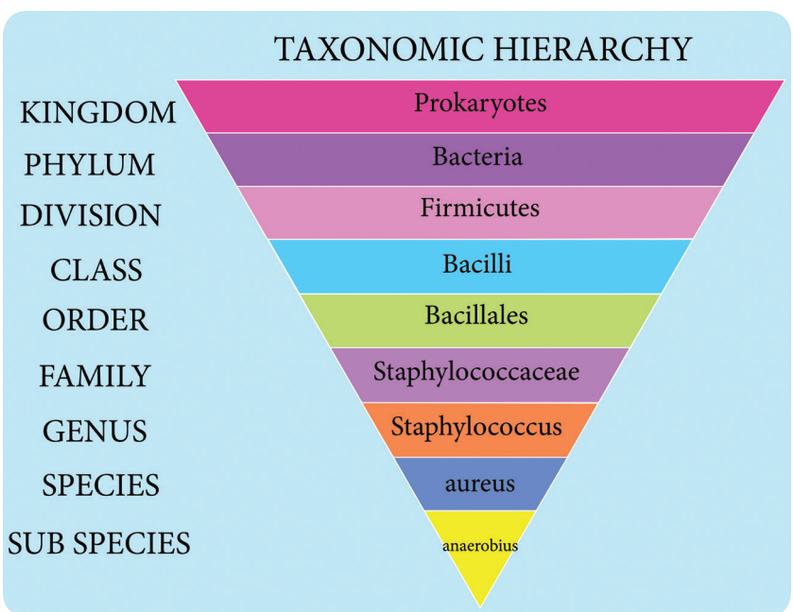


Figure 8.2: Taxonomic hierarchy-an example of hierarchy in microbial taxonomy

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